Using Feature Engineering and LSTM Networks to Detect Phishing Attacks and Homoglyph Obfuscation

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Introduction

This project proposes a machine learning-based system for detecting phishing emails and homoglyph attacks by analyzing email subject lines and identifying deceptive character patterns.

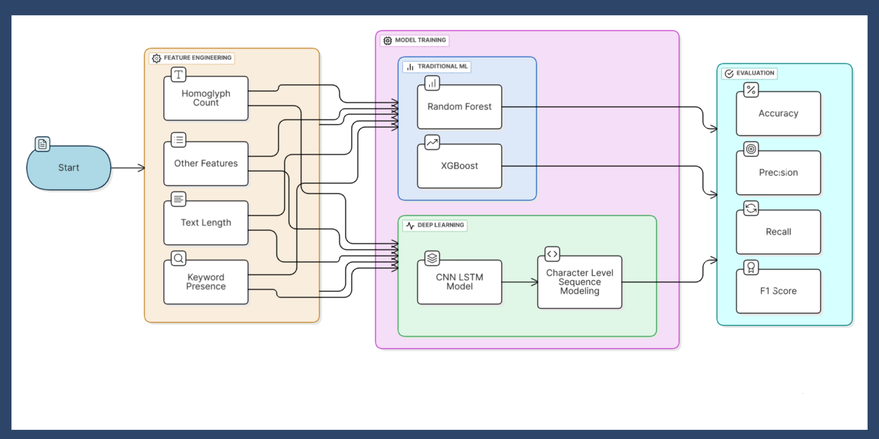
Objectives

- Curate phishing and benign datasets.
- Feature engineering for homoglyph detection.
- Train ML models: RF, XGBoost.
- Train DL model: LSTM.
- Evaluate and compare results.

Literature Review

Study	Findings	Research Gap
Thakur et al.	Used custom characteristics to identify phishing	No focus on homoglyph substitution or adversarial email headers
Almuhaideb et al.	Developed a Random Forest model for homoglyph detection in URLs	Approach did not extend to email content or phishing subjects
Lee et al.	Introduced PhishGAN, a generative model that simulates homoglyph attacks	No evaluation metrics for phishing classification performance
Altwaijry et al.	Benchmarked CNN, LSTM, and BERT models for phishing email detection	Did not integrate homoglyph-specific features

Methodology

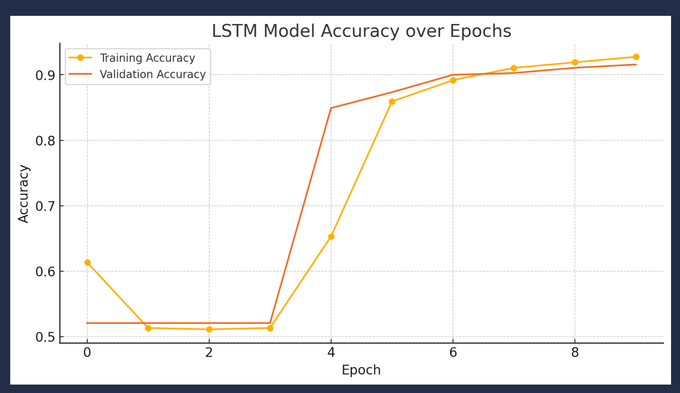


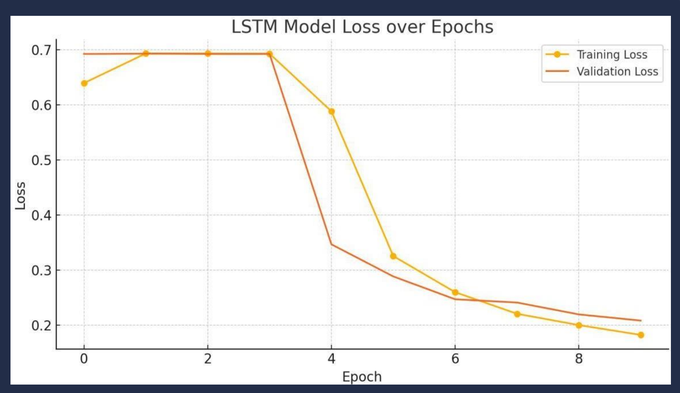
DATASET SOURCES

- Enron, SpamAssassin, CEAS, Ling-Spam
- Nazario, Nigerian Scam, Phishing Corpus
- Total ~80,000 samples, labeled 0 (Legit) / 1 (Phishing)

RESULTS & ANALYSIS

- Random Forest Accuracy: 75.37%
- XGBoostAccuracy: 75.00%
- CNN + LSTM Accuracy: 92.09%





- LSTM model captured sequential patterns more effectively
- Significant improvement in homoglyph detection

Model Comparison

Category	Model	Test Accuracy
Existing Method	Random Forest Classifier	75.37%
Existing Method	XGBoost Classifier	75.00%
Our Hybrid Training Model	CNN + LSTM (Deep Learning)	92.09%

Conclusion

- LSTM significantly outperformed traditional models in phishing detection
- Homoglyph-based obfuscations effectively detected with character-level analysis
- Machine learning + domain knowledge improves phishing email classification
- Future scope includes BERT integration and real-time deployment

Limitations

- Dataset imbalance and underrepresentation of zero-day homoglyph attacks.
- Homoglyph detection is limited by predefined character mappings.
- Lack of context from full email body or metadata.